

- Highest power density in SIP-8 metal package (optional plastic package)
- Wide 2:1 input voltage range
- Temperature range -40° to $+85^{\circ}\text{C}$
- High efficiency up to 89%
- Indefinite short-circuit protection
- I/O isolation 1600 VDC
- Remote On/Off control
- Fully RoHS compliant
- 3-year product warranty



The TMR 9 series is a new family of isolated 9W DC/DC converter modules with regulated output, featuring wide 2:1 input voltage ranges. The product comes in a ultra-compact SIP-8 metal package with a small footprint occupying only 2.0 cm² (0.3 square inch) of board space.

An excellent efficiency allows -40° to $+60^{\circ}\text{C}$ operation temperatures without derating. Further features include remote On/Off control and continuous short circuit protection. The very compact dimensions of these converters make them an ideal solution for many space critical applications in communication equipment, instrumentation and industrial electronics.

Models

Order Code	Input Voltage Range	Output 1		Output 2		Efficiency typ.
		Vnom	I _{max}	Vnom	I _{max}	
TMR 9-1210	9 - 18 VDC (12 VDC nom.)	3.3 VDC	2'000 mA			81 %
TMR 9-1211		5 VDC	1'600 mA			85 %
TMR 9-1219		9 VDC	1'000 mA			87 %
TMR 9-1212		12 VDC	750 mA			88 %
TMR 9-1213		15 VDC	600 mA			89 %
TMR 9-1215		24 VDC	375 mA			89 %
TMR 9-1221		+5 VDC	800 mA	-5 VDC	800 mA	85 %
TMR 9-1222		+12 VDC	375 mA	-12 VDC	375 mA	88 %
TMR 9-1223		+15 VDC	300 mA	-15 VDC	300 mA	89 %
TMR 9-2410	18 - 36 VDC (24 VDC nom.)	3.3 VDC	2'000 mA			82 %
TMR 9-2411		5 VDC	1'600 mA			85 %
TMR 9-2419		9 VDC	1'000 mA			88 %
TMR 9-2412		12 VDC	750 mA			89 %
TMR 9-2413		15 VDC	600 mA			90 %
TMR 9-2415		24 VDC	375 mA			90 %
TMR 9-2421		+5 VDC	800 mA	-5 VDC	800 mA	86 %
TMR 9-2422		+12 VDC	375 mA	-12 VDC	375 mA	89 %
TMR 9-2423		+15 VDC	300 mA	-15 VDC	300 mA	87 %
TMR 9-4810	36 - 75 VDC (48 VDC nom.)	3.3 VDC	2'000 mA			82 %
TMR 9-4811		5 VDC	1'600 mA			85 %
TMR 9-4819		9 VDC	1'000 mA			88 %
TMR 9-4812		12 VDC	750 mA			89 %
TMR 9-4813		15 VDC	600 mA			89 %
TMR 9-4815		24 VDC	375 mA			89 %
TMR 9-4821		+5 VDC	800 mA	-5 VDC	800 mA	86 %
TMR 9-4822		+12 VDC	375 mA	-12 VDC	375 mA	87 %
TMR 9-4823		+15 VDC	300 mA	-15 VDC	300 mA	87 %

Options

on demand (backorder with MOQ non stocking item)	- Optional models with plastic case
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Input Specifications

Input Current	- At no load	12 Vin models: 11 mA typ. 24 Vin models: 7 mA typ. 48 Vin models: 3 mA typ.
Surge Voltage		12 Vin models: 36 VDC max. (1 s max.) 24 Vin models: 50 VDC max. (1 s max.) 48 Vin models: 100 VDC max. (1 s max.)
Recommended Input Fuse		12 Vin models: 3'150 mA (slow blow) 24 Vin models: 2'500 mA (slow blow) 48 Vin models: 1'250 mA (slow blow) (The need of an external fuse has to be assessed in the final application.)
Input Filter		Internal Capacitor

Output Specifications

Voltage Set Accuracy		±1% max.
Regulation	- Input Variation (Vmin - Vmax) - Load Variation (0 - 100%) - Cross Regulation (25% / 100% asym. load)	single output models: 0.2% max. dual output models: 0.2% max. single output models: 1% max. dual output models: 1% max. (Output 1) 1% max. (Output 2) dual output models: 5% max.
Ripple and Noise (20 MHz Bandwidth)	- single output - dual output	3.3 Vout models: 50 mVp-p typ. (w/ 1 μ F X7R) 5 Vout models: 50 mVp-p typ. (w/ 1 μ F X7R) 9 Vout models: 50 mVp-p typ. (w/ 1 μ F X7R) 12 Vout models: 75 mVp-p typ. (w/ 1 μ F X7R) 15 Vout models: 75 mVp-p typ. (w/ 1 μ F X7R) 24 Vout models: 75 mVp-p typ. (w/ 1 μ F X7R) 5 / -5 Vout models: 50 / 50 mVp-p typ. (w/ 1 μ F X7R) 12 / -12 Vout models: 75 / 75 mVp-p typ. (w/ 1 μ F X7R) 15 / -15 Vout models: 75 / 75 mVp-p typ. (w/ 1 μ F X7R)
Capacitive Load	- single output - dual output	3.3 Vout models: 2'600 μF max. 5 Vout models: 1'300 μF max. 9 Vout models: 800 μF max. 12 Vout models: 560 μF max. 15 Vout models: 560 μF max. 24 Vout models: 200 μF max. 5 / -5 Vout models: 800 / 800 μF max. 12 / -12 Vout models: 390 / 390 μF max. 15 / -15 Vout models: 200 / 200 μF max.
Minimum Load		Not required
Temperature Coefficient		±0.02 %/K max.
Start-up Time		50 ms typ.
Short Circuit Protection		Continuous, Automatic recovery
Output Current Limitation		180% typ. of Iout max.
Transient Response	- Response Time	250 μs typ. (25% Load Step)

Safety Specifications

Standards	- IT / Multimedia Equipment	EN 60950-1 EN 62368-1 IEC 60950-1 IEC 62368-1 UL 60950-1 UL 62368-1
	- Certification Documents	www.tracopower.com/tmr9-safety-cert/

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

Energy Source	- Output, acc. to 62368-1	ES2
Power Source	- Output, acc. to 62368-1	PS3
Pollution Degree		PD 2
Over Voltage Category		Not mains connected

EMC Specifications

EMI (Emissions)	- Conducted Emissions	EN 55032 class A (with external filter)
	- Radiated Emissions	EN 55032 class B (with external filter)
External filter proposal:		www.tracopower.com/tmr9-emc-filter/
EMS (Immunity)	- Electrostatic Discharge	Air: EN 61000-4-2, ± 8 kV, perf. criteria A
		Contact: EN 61000-4-2, ± 6 kV, perf. criteria A
	- RF Electromagnetic Field	EN 61000-4-3, 20 V/m, perf. criteria A
	- EFT (Burst) / Surge	EN 61000-4-4, ± 2 kV, perf. criteria A
		EN 61000-4-5, ± 2 kV, perf. criteria A
	- Conducted RF Disturbances	Ext. input component: 24 Vin models: KY 220 μ F SMDJ70A
	- PF Magnetic Field	48 Vin models: KY 220 μ F SMDJ120A
		Continuous: EN 61000-4-6, 10 Vrms, perf. criteria A
		EN 61000-4-8, 100 A/m, perf. criteria A

General Specifications

Relative Humidity		95% max. (non condensing)
Temperature Ranges	- Operating Temperature	-40°C to +85°C
	- Case Temperature	+100°C max.
	- Storage Temperature	-55°C to +125°C
Power Derating	- High Temperature	Depending on model
		See application note: www.tracopower.com/tmr9-cc/
Cooling System		Natural convection (20 LFM)
Remote Control	- Voltage Controlled Remote (passive = on)	On: 0 to 0.5 VDC or open circuit
		Off: 3 to 12 VDC
	- Off Idle Input Current	Refers to 'Remote' and '-Vin' Pin
	- Remote Pin Input Current	2.5 mA typ.
Altitude During Operation		0.5 to 2.5 mA
Regulator Topology		5'000 m max.
Switching Frequency		Flyback Converter
Insulation System		400 kHz typ. (PWM) (single output models)
		500 kHz typ. (PWM) (dual output models)
Isolation Test Voltage	- Input to Output, 60 s	Functional Insulation
	- Input to Case, 60 s	1'600 VDC
Isolation Resistance	- Input to Output, 500 VDC	1'000 VDC
Isolation Capacitance	- Input to Output, 100 kHz, 1 V	1'000 M Ω min.
Reliability	- Calculated MTBF	50 pF max.
		2'940'000 h (for standard version)
		2'700'000 h (for plastic version)
Washing Process		(MIL-HDBK-217F, ground benign)
		According to Cleaning Guideline
Environment	- Vibration	www.tracopower.com/info/cleaning.pdf
	- Thermal Shock	MIL-STD-810F
Housing Material		MIL-STD-810F
Potting Material		Copper (for standard version)
Pin Material		Non-conductive plastic (for plastic version)
Pin Foundation Plating		Silicone (UL 94 V-0 rated)
Pin Surface Plating		Copper
		Nickel (2 - 3 μ m)
		Tin (3 - 5 μ m), matte

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

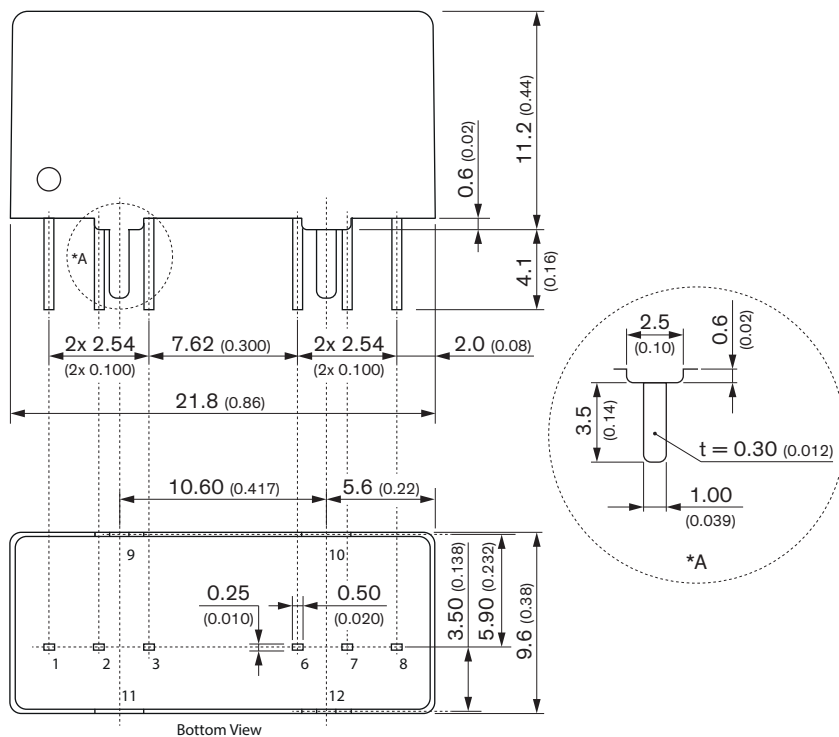
Housing Type	Metal Case
Mounting Type	PCB Mount
Connection Type	THD (Through-Hole Device)
Footprint Type	SIP8
Soldering Profile	Lead-Free Wave Soldering 260°C / 6 s max.
Weight	5.9 g (for standard version) 4.8 g (for plastic version)
Environmental Compliance	- REACH Declaration www.tracopower.com/info/reach-declaration.pdf REACH SVHC list compliant REACH Annex XVII compliant www.tracopower.com/info/rohs-declaration.pdf Exemptions: 7(a), 7(c)-I (RoHS exemptions refer to the component concentration only, not to the overall concentration in the product (O5A rule).) a0c3442b-7838-4ff2-befa-e4362c1436dc
	- RoHS Declaration
	- SCIP Reference Number

Additional Information

Supporting Documents	www.tracopower.com/overview/tmr9
Frequently Asked Questions	www.tracopower.com/glossary-faq
Glossary	www.tracopower.com/info/glossary.pdf
Service Portal	www.tracoserviceportal.com

Outline Dimensions

Metal package (standard)



Pinout		
Pin	Single Output	Dual Output
1	-Vin (GND)	-Vin (GND)
2	+Vin (Vcc)	+Vin (Vcc)
3	Remote On/Off	Remote On/Off
6	+Vout	+Vout
7	-Vout	Common
8	NC	-Vout
9	Case*	Case*
10	Stand off	Stand off
11	Stand off	Stand off
12	Case*	Case*

NC: Not connected

*Case pins must not be connected to any circuit.

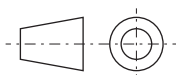
Avoid routing PCB traces under the converter.

Dimensions in mm (inch)

Tolerances: x.x ±0.5 (x.xx ±0.02)

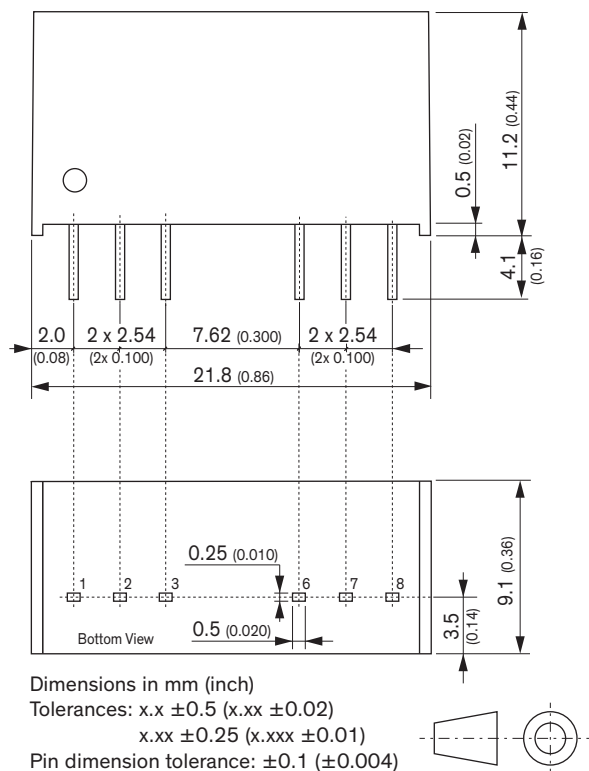
x.xx ±0.25 (x.xxx ±0.01)

Pin dimension tolerance: ±0.1 (±0.004)



All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

Plastic package (option)



Pinout		
Pin	Single Output	Dual Output
1	–Vin (GND)	–Vin (GND)
2	+Vin (Vcc)	+Vin (Vcc)
3	Remote On/Off	Remote On/Off
6	+Vout	+Vout
7	–Vout	Common
8	NC	–Vout

NC: Not connected

Avoid routing PCB traces under the converter.